C S Vaishakh

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Professional Summary

AI & Data Science undergraduate focused on backend development and systems architecture. Builds and deploys REST APIs in Go (Fiber, Gin) and Node.js (Express) over PostgreSQL and Redis.

Education

St. Joseph's College of Engineering and Technology, Palai Kottayam, Kerala
B.Tech in Artificial Intelligence and Data Science — APJ Abdul Kalam Technological University *2023 – 2027*

Projects

Quickhand December 2025 – Present

- Implemented role-based access control across two user roles, securing Fiber v3 routes with stateless JWTs.
- Built an authenticated WebSocket layer broadcasting real-time job alerts to matched handymen.
- Structured the codebase as a Go workspace monorepo pairing the Fiber API with a TanStack Start frontend.
- **Technologies**: Go, Fiber v3, WebSockets, React, TanStack Start, TypeScript, PostgreSQL, GORM, Docker

Snip July – August 2026

- Built a URL shortener with a Go (Gin) REST API, React dashboard, JWT auth, and collision-safe codes.
- Cut redirect latency by fronting PostgreSQL with a Redis cache on the public redirect path.
- Deployed the frontend on Vercel and the backend on AWS EC2 behind Caddy via GitHub Actions CI/CD.
- **Technologies**: Go, Gin, GORM, PostgreSQL, Redis, JWT, React, Vite, Docker, AWS EC2, Vercel

PortfolyoIQ March – August 2026

- Engineered a privacy-preserving rebalancing platform: Next.js client, Express API, FastAPI model service.
- Trained logistic regression models in-browser with TensorFlow.js, eliminating raw portfolio data uploads.
- Implemented Federated Averaging (FedAvg) on the Express aggregator to synchronize local model updates.
- Self-hosted all three services on AWS EC2 behind Caddy via systemd, deployed by GitHub Actions.
- **Technologies**: Next.js, Express.js, FastAPI, TensorFlow.js, Drizzle ORM, PostgreSQL, Bun, AWS EC2

ONT Monitor April 2026

- Built a Go microservice (Fiber v3, GORM) ingesting real-time ONT status change events.
- Implemented thread-safe in-memory queues and a worker pool to evaluate outages without race conditions.
- Integrated the Twilio API to dispatch outage and recovery WhatsApp alerts on state transitions.
- **Technologies**: Go, Fiber v3, GORM, PostgreSQL, Twilio API, Docker

Technical Skills

Programming: Python, TypeScript, Go, Java, JavaScript
Web & Backend: Next.js, React, Express.js, Fiber v3, Gin, FastAPI, Node.js, REST APIs, WebSockets
Databases & ORMs: PostgreSQL, MongoDB, Redis, GORM, Drizzle ORM, Mongoose
AI / ML: TensorFlow.js, scikit-learn, NumPy, Pandas, Feature Engineering, Model Evaluation
Tools & DevOps: Git, GitHub Actions (CI/CD), Docker, AWS EC2, Caddy, Vercel, Bun, Linux, Postman